

HDOC DAY-13 ACTIVITY

QUESTION 1 – Menu-Driven Digit Operations

Write a menu-driven C program using switch-case to input a positive integer and perform one of the following operations based on the user's choice:

- Count the number of even digits, and odd digits. E.g. Number: 12045, Even_dig = 3, Odd_dig = 2
- Find the sum of its even digits and the sum of its odd digits separately. E.g. Number: 12345, Sum_even = 6, Sum_odd = 9

1. Problem Understanding

- Input: A positive integer and the user's menu choice.
- For the second operation, no additional input is required.
- Output: The result of the operation selected through switch-case.

Operation Summary

Choice	Operation	Main Logic
1	Count even and odd digits	Extract each digit and count whether it is even or odd.
2	Sum even and odd digits separately	Extract each digit and add it to the appropriate sum.
Other	Invalid choice	Display an appropriate message.

2. Standard Test Cases

Test Case	Choice	Input	Expected Output	Actual Output	Result
1	1	n = 12045	Even_dig = 3; Odd_dig = 2	Even_dig = 3; Odd_dig = 2	Pass
2	1	n = 2468	Even_dig = 4; Odd_dig = 0	Even_dig = 4; Odd_dig = 0	Pass
3	2	n = 12345	Sum_even = 6; Sum_odd = 9	Sum_even = 6; Sum_odd = 9	Pass
4	2	n = 13579	Sum_even = 0; Sum_odd = 25	Sum_even = 0; Sum_odd = 25	Pass

3. Algorithm

A. Common Menu Logic

1. Start the program.
2. Declare the required variables.
3. Read the positive integer.
4. Display the menu and read the user's choice.
5. Use switch-case to perform the selected operation.
6. Display the result.
7. Stop the program.

B. Case 1 – Count Even and Odd Digits

8. Initialize even_dig and odd_dig to zero.
9. Extract the last digit using $n \% 10$.
10. Check whether the digit is divisible by 2.
11. If it is divisible by 2, increment even_dig; otherwise increment odd_dig.
12. Remove the last digit using $n / 10$ and repeat until all digits are processed.
13. Display the number of even and odd digits.

C. Case 2 – Sum Even and Odd Digits

14. Initialize sum_even and sum_odd to zero.
15. Extract the last digit using $n \% 10$.
16. If the digit is divisible by 2, add it to sum_even; otherwise add it to sum_odd.
17. Remove the last digit using $n / 10$ and repeat until all digits are processed.
18. Display sum_even and sum_odd.

5. C Program

```
#include <stdio.h>
```

```
int main(void)
```

```
{
```

```
int n, choice, digit;
```

```
int even_dig, odd_dig;
```

```
int sum_even, sum_odd;
```

```
int temp;
```

```
printf("Enter a positive integer: ");
```

```
scanf("%d", &n);
```

```
printf("\nMenu\n");
```

```
printf("1. Count even and odd digits\n");

printf("2. Find sum of even and odd digits separately\n");

printf("Enter your choice: ");

scanf("%d", &choice);

switch (choice)

{

case 1:

even_dig = 0;

odd_dig = 0;

temp = n;

while (temp > 0)

{

digit = temp % 10;

if (digit % 2 == 0)

even_dig++;

else

odd_dig++;

temp = temp / 10;

}

printf("Even_dig = %d\n", even_dig);

printf("Odd_dig = %d\n", odd_dig);

break;

case 2:

sum_even = 0;

sum_odd = 0;

temp = n;

while (temp > 0)
```

```

{

digit = temp % 10;

if (digit % 2 == 0)

sum_even = sum_even + digit;

else

sum_odd = sum_odd + digit;

temp = temp / 10;

}

printf("Sum_even = %d\n", sum_even);

printf("Sum_odd = %d\n", sum_odd);

break;

default:

printf("Invalid choice.\n");

}

return 0;

}

```

6. Compilation and Execution Evidence

```

kali@kali: ~/Desktop
Session Actions Edit View Help
(kali@kali)~/Desktop
$ gedit question.c
** (gedit:18606): WARNING **: 11:50:01.821: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found
** (gedit:18606): WARNING **: 11:50:01.821: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found
(kali@kali)~/Desktop
$ gcc question.c -o question
(kali@kali)~/Desktop
$ ./question
Enter a positive integer: 12045

Menu
1. Count even and odd digits
2. Find sum of even and odd digits separately
Enter your choice: 1
Even_dig = 3
Odd_dig = 2
(kali@kali)~/Desktop
$ ./question
Enter a positive integer: 2468

Menu
1. Count even and odd digits
2. Find sum of even and odd digits separately
Enter your choice: 1
Even_dig = 4
Odd_dig = 0
(kali@kali)~/Desktop
$ ./question
Enter a positive integer: 12345

Menu
1. Count even and odd digits
2. Find sum of even and odd digits separately
Enter your choice: 2

```

```
Session Actions Edit View Help
1. Count even and odd digits
2. Find sum of even and odd digits separately
Enter your choice: 1
Even_dig = 3
Odd_dig = 2

(kali@kali)-[~/Desktop]
└─$ ./question
Enter a positive integer: 2468

Menu
1. Count even and odd digits
2. Find sum of even and odd digits separately
Enter your choice: 1
Even_dig = 4
Odd_dig = 0

(kali@kali)-[~/Desktop]
└─$ ./question
Enter a positive integer: 12345

Menu
1. Count even and odd digits
2. Find sum of even and odd digits separately
Enter your choice: 2
Sum_even = 6
Sum_odd = 9

(kali@kali)-[~/Desktop]
└─$ ./question
Enter a positive integer: 13579

Menu
1. Count even and odd digits
2. Find sum of even and odd digits separately
Enter your choice: 2
Sum_even = 0
Sum_odd = 25

(kali@kali)-[~/Desktop]
└─$
```

7. Verification of All Test Cases

Test Case	Choice	Input	Expected Output	Actual Output	Result
1	1	n = 12045	Even_dig = 3; Odd_dig = 2	Even_dig = 3; Odd_dig = 2	Pass
2	1	n = 2468	Even_dig = 4; Odd_dig = 0	Even_dig = 4; Odd_dig = 0	Pass
3	2	n = 12345	Sum_even = 6; Sum_odd = 9	Sum_even = 6; Sum_odd = 9	Pass
4	2	n = 13579	Sum_even = 0; Sum_odd = 25	Sum_even = 0; Sum_odd = 25	Pass